Module 1 Homework 3 Challenge

Below is a scored review of your assessment. All questions are shown.

Earned 89,991 of 99,990 points. 90%

Main Question Set

Solve six of the first ten questions to achieve a passing score of 60%. Don't miss the learning opportunity in the rest of the problems, though -- spend the full time and attempt every one of them!

Correct Answer Partially Correct Incorrect Answer

1 A metal alloy is 20% silver. How many grams of silver are found in 80 grams of alloy?

Your Answer	16
Correct Answer	16

Explanation

Silver is 20% of the alloy, so it is $0.2 \times \text{ total alloy weight} = 0.2 \times 80 \text{ grams} = 16$ grams.

2

Suppose you invested in some unusual stock which goes up in price by 10% whenever the year is odd, and then drops by 10% whenever the year is even. If you keep this stock for 20 years, what will be the change in the value of your investment over this time?

Your Answer	The stock will decrease overall over the 20 years
Correct Answer	The stock will decrease overall over the 20 years

Explanation

We saw in the lesson that if you look at two consecutive years, a $+10\%$ change followed by a -10% change will result in an overall loss in price. This is because after the initial price increase of $+10\%$, the current price will be higher than the original price that you purchased the stock at. Then, the amount that the stock drops by after the -10% change is actually bigger than what you added in the first year. The price graph would look like a staircase, going up at first and then down, with an overall downward trajectory. The correct answer is the second choice.

3

Alice, Bob and Charlie went to the movies. Alice contributed $\$15$ to the cost of the tickets, Bob contributed $\$6$ to the cost of the tickets, and Charlie forgot his money, so he contributed nothing. How would you find the amount of money that Charlie owes Alice and Bob?

Your Answer

Each person's ticket costs $\frac{\$6+\$15}{3} = \frac{\$21}{3} = \$7$, so Charlie should pay Alice $\$7$, but he shouldn't pay Bob anything, because Bob actually underpaid for his ticket, also. Bob should pay Alice an additional $\$1$ to make up for the fact that he only paid $\$6$ at the beginning.

Correct Answer

Each person's ticket costs $\frac{\$6+\$15}{3} = \frac{\$21}{3} = \$7$, so Charlie should pay Alice $\$7$, but he shouldn't pay Bob anything, because Bob actually underpaid for his ticket, also. Bob should pay Alice an additional $\$1$ to make up for the fact that he only paid $\$6$ at the beginning.

Explanation

The second choice is incorrect because it doesn't take into account that Alice and Bob already paid money, so they shouldn't have to pay $\$7$ again.

The third choice is wrong because in this situation, Bob will be paying for everyone's tickets, and Bob and Alice will end up getting money rather than paying money -- furthermore, Bob will receive more net money than Alice.

The fourth choice is wrong because Charlie is paying more than the cost of a single ticket, $\$7$, to Alice.

The first choice is correct because it takes into account how much money each person already paid, and the fact that in the end, each should have paid $\$7$.

4 There are cats, dogs and pigs in a ratio of $12:13:15$, and there are 24 more pigs than cats. How many animals are there altogether?

Your Answer	320
Correct Answer	320

Explanation

Using the ratio of the cats, dogs, and pigs, we can say that there are $12n$ cats, $13n$ dogs and $15n$ pigs. Then there are $15n - 12n = 3n$ more pigs than cats. Setting this equal to 24 means that $n = 8$. The total number of animals is $12n + 13n + 15n = 40n$, so the number of total animals is $8 \times 40 = 320$.

5

A koala is sleeping on the top of a hill. Every minute, the koala slips 3 inches down. In 18 minutes, the koala will be halfway down the hill. How many inches tall is the hill? Your answer should be the number only, without units.

✓	**Your Answer**	108
	Correct Answer	108

Explanation

Explanation:

In 18 minutes, the koala will have slipped down $3 \times 18 = 54$ inches. Since this is halfway, the hill is twice this length, or $2 \times 54 = 108$.

6 Suppose that all of these gears are all connected by a chain to a driving motor. Which of the following gears will spin at the fastest RPM (revolutions per minute)?

Your Answer	A gear with 10 teeth
Correct Answer	A gear with 10 teeth

Explanation

Gears spin as their teeth are gripped by some corresponding chain or gear that meshes with them. We can think of the number of teeth that are moved per unit time, and that translates to the revolutions per minute rotational speed of the gear. The number of teeth that need to be moved per minute for the gear with only 10 teeth is low, so that gear moves the fastest.

7 Write the number 2946573833993 in scientific notation.

Your Answer	2.9×10^{12}
Correct Answer	2.9×10^{12}

Explanation

Do you remember the rules for scientific notation? This number has 13 digits, so we know that we will be multiplying by 10^{12}. The number we multiply by will need to be between 1 and 10, so we get 2.9×10^{12}.

8 Alex designs a model of the Empire State Building that is 4 inches wide and 10 inches high. If the real Empire State Building is 1250 feet tall, how wide, in feet, is the Empire State Building according to Alex's proportions?

Your Answer	500
Correct Answer	500

Explanation

According to Alex's dimensions, the Empire State Building's width to height ratio is $\frac{2}{5}$. Therefore, its width is

$$\begin{aligned}\frac{2}{5} \times 1250 &= 2 \times 250 \\ &= 500\end{aligned}$$

9 In order to transport goods, there are 16 vans total, and each must make 12 trips. How many vans will be needed in order to transport the same amount of cargo, if the number of trips that each van takes is increased by one-third ($\frac{1}{3}$)?

Your Answer	12
Correct Answer	12

Explanation

The total number of trips that need to be made in order to transport all the goods is

$$16 \times 12 = 192 \text{ trips}$$

After increasing the number of trips by a third, the number of trips that one vans takes will be equal to

$$12 + \frac{1}{3} \times 12 = 16 \text{ trips}$$

So, the number of vans that will be needed in order to transport the same amount of cargo is equal to

$$\frac{192}{16} = 12 \text{ vans}$$

10 The A Company charges an initial cost of $\$250$, plus $\$75$, for every month you rent out an apartment. The B Company charges an initial cost of $\$375$, plus $\$70$, for every month that you rent out the same building. For what number of months are both companies A and B charging you the same amount overall?

Your Answer	[no answer given]
Correct Answer	25

Explanation

Let's let m be the number of months that you rent the apartment. Then, using the information, we can write a simple equation that equates the total cost of renting an apartment from Company A for m months to the total cost of renting an apartment from Company B for m months:

$$\begin{aligned} 250 + 7m &= 375 + 70m \\ 125 &= 5m \\ 25 &= m \\ &= 25 \end{aligned}$$

Earned 0 of 7 points. 0%

Bonus Challenge Questions

These optional questions will not count towards your final score. However, we strongly suggest trying them for the benefit of the challenge!

11 You already have one cent in your piggy bank. Your boss gives you 1 cent on day one, 2 cents on day two, 4 cents on day three, 8 cents on day four, and so on. All payments are in pennies. Your piggy bank will become full after the payment on the ninth day. After which payment will your piggy bank be half full?

Your Answer	[no answer given]
Correct Answer	8

Explanation

The key observation to make here is that you are doubling your money every payment, and if your bank becomes full on the ninth payment, it must have been half full on the eigth payment.

12 Sixteen pears are equal in cost to 20 apples, and 25 apples are equal in cost to 35 bananas, and 63 bananas are equal in cost to $\$3$. How many dollars do 12 pears cost? Your answer should be a number only, without units.

Your Answer	[no answer given]
Correct Answer	1

Explanation

We are given a ratio of 16 pears to 20 apples, which simplifies to 4 pears being equivalent to 5 apples. Let's find the number of dollars that 4 pears costs, since finding the cost of 1 pear would give us intermediate fractions. The number 4 makes sense because it is just $\frac{1}{3}$ of what we want, which is 12 pears.

The ratio of apples to bananas is $25:35$, which simplifies to $5:7$, or $1:\frac{7}{5}$. We can look at this ratio and see that it matches well with the number of apples, which is 5. Choosing numbers that make the calculations simple is a neat way to do this problem! Then it's easy to see that the 5 apples are equivalent to the 7 bananas. The 7 bananas are $\frac{1}{9}^{\text{th}}$ of the cost of the 63 bananas, which is $\$3$, so we can divide this by 9 to get the cost of the original 4 pears:

$$\frac{\$3}{9} = \$\frac{1}{3}$$

Don't forget that we actually want the cost of $\$12$ pears, so we should multiply by 3, to get $\$\frac{1}{3} \times 3 = \$1$.

13

Open-topped cubical cardboard boxes are manufactured with costs directly proportional to their surface areas. A $3 \times 3 \times 3$ box costs $\$3.24$. How many dollars does a $5 \times 5 \times 5$ box cost to make? Your answer should be the number only, without units.

✕	**Your Answer**	[no answer given]
	Correct Answer	9

Explanation

The surface area of a box is related to the side length of the box squared, so the ratio of the larger box's surface area to the smaller box's surface area would be

$$\frac{5^2}{3^2} = \frac{25}{9}.$$

We don't even have to calculate the cost per unit surface area; we can just use this ratio to figure out how much the cost changes when we go from using the smaller box to the bigger box:

$$\$3.24 \times \frac{25}{9}$$

Notice that you can divide $\$3.24$ nicely by 9, which you can see from adding up the digits of $\$3.24$ and seeing that they add up to a multiple of 9. So we can take the shortcut of doing

$$\frac{\$3.24}{9} \times 25 = \$0.36 \times 25$$

Every group of four 25's gives us a 100, so $36 \times 25 = 9 \times 100$, and since we have $\$0.36$ instead of $\$36$, divide by 100 to get $\$9$ as our answer.

14 Bill usually makes chicken soup for his family of four, using 7 cups of broth, 3 tablespoons of salt, 2.5 cups of vegetables, and 1 teaspooon of basil. Today he is using the same recipe, but only making one portion for himself. If c is the number of cups of broth, s is the number of tablespoons of salt, v is the number of cups of vegetables, and b is the number of teaspoons of basil he will need, what is $c+s+v+b$, expressed as a simplified fraction? Your answer should be the fraction only, without spaces, e.g. 1/2

✕	**Your Answer**	[no answer given]
	Correct Answer	27/8

Explanation

The amount that Bill usually makes, for four people, is

$$\begin{aligned} 4c+4s+4v+4b &= 7+3+2.5+1 \\ &= 13.5 \end{aligned}$$

because this is the amount that Bill usually makes for four people. Then, the amount that he makes for himself is

$$\begin{aligned} c+s+v+b &= \frac{13.5}{4} \\ &= \frac{27}{2} \times \frac{1}{4} \\ &= \frac{27}{8} \end{aligned}$$

15 In a family, each boy has one more brother than sister. Each girl has 2 times as many brothers as sisters. How many total children are there in the family?

Your Answer	[no answer given]
Correct Answer	10

Explanation

Let's let the number of boys equal b, and the number of girls equal g. From the first sentence, which says that each boy has one more brother than sister, we can write

$$\begin{aligned} b - 1 &= 1 + g \\ b &= 2 + g \end{aligned}$$

From the second statement, which says that each girl has two times as many brothers as sisters, we can write

$$\begin{aligned} 2(g - 1) &= b \\ 2g - 2 &= 2 + g \\ g &= 4 \end{aligned}$$

Then, $b = 2 + g = 2 + 4 = 6$, and the number of total children is the sum of the number of boys and the number of girls, which is $4 + 6 = 10$.

16

Everyone invests certain amount, based on returns divided in investment proportion how much should each person get back Alice, Bob and Charlie each invest $\$100, \$200$ and $\$300$ into an item worth $\$600$. Suppose the value of the item grows to $\$900$, and the three people decide to cash out; if they divide the money fairly (i.e. based on the proportions of the initial investment), how much profit did Bob make, in dollars?

Your Answer	[no answer given]
Correct Answer	100

Explanation

Alice should obtain $\frac{1}{6}$ of the growth , Bob should obtain $\frac{2}{6} = \frac{1}{3}$ of the growth, and Charlie should obtain $\frac{3}{6} = \frac{1}{2}$ of the growth. The growth is $900 - 600 = 300$ so Bob obtains $\frac{300}{3} = 100$.

17 Brand A says they offer 25% more juice than Brand B for only 90% of the cost. What fraction of the unit price that Brand B offers for its orange juice is the unit price that Brand A offers for its orange juice? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2

Your Answer	[no answer given]
Correct Answer	18/25

Explanation

It might be easier to calculate the inverse fraction, the fraction of the amount of orange juice that Brand A offers, as compared to the amount of orange juice that Brand B offers. Then, at the end, we can flip the fraction to get its reciprocal, which means the fraction that Brand A's orange juice costs, compared to Brand B.

Brand A offered 25% more orange juice than Brand B for the same cost. Then, the volume of Brand A's orange juice box would be $(1 + 0.25)$ times the volume of Brand B's orange juice box. However, since Brand A is cheaper, since its orange juice costs 90% of Brand B, we should multiply this ratio by $\frac{10}{9}$.

Thus, the amount of orange juice that Brand A offers for 100% of the cost that Brand B offers for one unit of orange juice is

$$\frac{10}{9} \times \frac{5}{4} = \frac{25}{18}$$

Therefore, the fraction that we want, which means the fraction of Brand B's cost that is Brand A's unit cost, is the reciprocal, which is

$$\frac{18}{25}$$

Check that this is in simplest form, and we are done!

Earned 0 of 3 points. 0%

Super Bonus Questions

These questions are an extreme challenge! Some of the most interesting, creative and clever questions are the hardest ones. So try every single one. Even if you can't figure it out, you'll still be deepening your understanding of the important math concepts that underlie this course.

18 A customer in a shoe store bought a pair of shoes on sale for $\$15$, paying with a $\$20$ bill. He didn't have change, so he asked a lady for change. She gave him a $\$10$ bill and two $\$5$ bills for the $\$20$ bill. The salesman gave the customer $\$5$ change. Then, the lady who just gave the change came and said that the $\$20$ bill was counterfeit. The salesman examined the $\$20$ bill and agreed that it was worthless, and gave the lady another $\$20$ bill. That evening, the shoe salesman was trying to reconcile his balance sheet for the day and he became very confused. What total value did the shoe store actually lose in this strange transaction? Your answer should be the number only, without units.

✕	**Your Answer**	[no answer given]
	Correct Answer	20

Explanation

The key is to keep a close track of where any value changed hands. The customer received the $\$15$ shoe and $\$5$ in change. In return, he gave a fake $\$20$ bill, which is worth $\$0$ in value. The salesman and lady break even, as the lady gives the salesman a $\$10$ bill and two $\$5$ bills. Though the salesman initially gave her the counterfeit bill, he does eventually give her a real $\$20$. However, none of that interaction changed what happened between the salesman and the customer, so the salesman has still lost the $\$15$ shoe and $\$5$ in change, which is a total of $\$20$ in losses.

19 Jamal makes pink paint by combining red paint and white paint in a ratio of $2:7$, respectively. If red paint and pink paint cost twice as much as white paint at the store, what fraction of the store pink paint's unit cost is Jamal's pink paint's unit cost? Express your answer as a simplified fraction, without spaces, e.g. 1/2

Your Answer	[no answer given]
Correct Answer	11/18

Explanation

This problem doesn't give the exact costs of each color of paint, and since the answer we want is a fraction of costs, this answer should be the same no matter what price we arbitrarily decide for the cost of the white paint.

Let's suppose that white paint costs $\$1$ per unit of paint. Then, since the red paint and pink paint are both twice as much as the white paint, they are $\$2$ per unit paint.

To make 9 units of Jamal's pink paint, it will cost

$$2(\$2) + 1(\$7) = \$11$$

dollars, since Jamal uses 2 units of red paint and 7 units of white paint.

Since pink paint is twice as expensive as white paint, in order to buy 9 units of the store's pink paint, this will cost $\$2 \times 9 = \$18$. We want the ratio of Jamal's cost to the store's cost, so this is $\frac{11}{18}$.

20 There are two 3-D printers, one faster and one slower. The slower printer's speed of printing is $\frac{3}{4}$ the speed of the faster one. The slower printer is printing a cube of side length 5, and the faster printer is printing a cube of side length 3. Consider how fast the height of each cube grows (note, not the volume, but the height.) What is the ratio of the height-growth rate of the 3 cube compared to the 5 cube? Your answer should be a fraction in simplest form, without spaces, e.g. 1/2.

Your Answer	[no answer given]
Correct Answer	100/27

Explanation

Imagine that the printers are outputting stacked squares to create the cubes. The fast printer has to output squares of area $3^2 = 9$ to create the cube with side length 3, and the slow printer has to output squares of area $5^2 = 25$ to create the cube with side length 5. As the cubes are being formed, their height changes based on the number of squares added. We know the faster printer outputs squares $\frac{4}{3}$ faster than the slower printer, so the desired height-growth ratio should be the speed ratio times the square area ratio, or

$$\frac{4}{3} \times \frac{25}{9} = \frac{100}{27}.$$

Summary

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